

CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Project Interim Report

**Smart Irrigation System Using IoT and
Machine Learning for Sustainable Agriculture in
Nepal**

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1 Introduction and Context

Agriculture is responsible for 70–80% of the world’s freshwater use. That means, enhancing irrigation is a key part of any plan for sustainable food production (Al-Mulla and Al-Badi, 2022; Liu et al., 2023). Recent developments in the Internet of Things (IoT), artificial intelligence (AI), and wireless sensor networks have made it possible for farms to make decisions based on data rather than relying solely on intuition and watering as a fixed schedule (Singha et al., 2025; Ikdid et al., 2021). Unfortunately, traditional methods are still inefficient in many places, resulting in the excessive watering problem, the lack of watering problem, and the problem of crop losses that could have been prevented (Naji and Salman, 2021; Rajpoot and Singh, 2022).

Nepal is a strong candidate for intervention. Around 60% of the labor force is engaged in agriculture. Yet water supplies are often disrupted by seasonal monsoon cycles, breakdowns in the water-related infrastructure, and limited rural electrification (Kunwar, 2024; Mool et al., 2001). These infrastructural barriers call for a technology that is locally suitable rather than a direct copy of solutions developed for well-resourced settings. As a result, this paper aims at creating a low-cost intelligent irrigation system for the Nepalese smallholder farmers. This system integrates the use of IoT sensors, edge computing, and machine learning.

This report is divided into different parts.

Section 2 explores the relevant literature and finds research gaps.

Section 3 specifies the project aims, objectives, and scope.

Section 4 describes the system design and evaluation system.

Section 5 considers feasibility, risks, and ethical issues.

1.1 Problem Statement

Nepalese smallholder farmers mainly base their irrigating schedule on their experience and tradition (Kunwar, 2024), which means the crops get watered based on a fixed routine and hardly the actual soil and weather conditions. As a result, there is a habitual wastage of water and yields are not optimized.

This project is an attempt to bridge this gap by using IoT sensors combined with a machine learning model trained on local conditions, which make possible everyday monitoring and automatic, data-driven irrigation management.

2 Literature Review

2.1 IoT-Based Smart Irrigation

Singha et al. (2025) created a system based on ESP32 with soil moisture, humidity, and temperature sensors, which led to a 12% increase in yield and a 58% decrease in water usage when compared with the traditional methods of growing red amaranth. Naji and Salman (2021)

designed a low-cost wireless sensor network with Arduino and XBee for soil monitoring and irrigation control. Rajpoot and Singh (2022) through a comparative study, revealed that a sensor-based system could much reduce human labor for soil monitoring and improve plant growth. Their results are mainly relevant for Nepal's Terai region because of its agro-climatic similarities with northern India.

On the community level, Ikdid et al. (2021) came up with a collaborative irrigation system employing the Penman-Monteith model was developed to supply water based on the real water requirements of crops; But, the assessment was conducted only in experimental conditions and there was no consideration of the smallholder farmers' willingness to adopt the technique. The provision of electricity in far-flung areas continues to be a major challenge: Sohan and Nahar (2022) overcame this issue by proposing a solar-powered, low-energy sensor network with a battery backup, an architecture that has since been used as a reference in this project.

2.2 Nepal-Specific Agricultural Context

International literature should be considered with the specific circumstances of Nepal. For example, Mool et al. (2001) outlined the exposure of Nepal's water resources to not only seasonal variations but also to the melting of glaciers. They pointed out that the decrease in water flow during the dry season will, in the future, limit irrigation. Upadhyay et al. (2015) in their research found that farmer-managed irrigation systems in Nepal's hill districts face problems of unfair distribution and insufficient measures to crop stress in dry spells. These research works are a clear indication that the problem is both technically solvable and time pressing in the local context.

2.3 Artificial Intelligence and Machine Learning

Meng et al. (2023) showed that LSTM (Long Short-Term Memory) networks can predict irrigation time-series more accurately than conventional recurrent architectures while running at twice the inference speed. Chen et al. (2021) stated that purely mathematical models cannot sufficiently represent complex soil-water-plant interactions that are nonlinear, which is, in fact, the rationale for the machine learning approach employed in the present work. Tang et al. (2023) designed a fuzzy logic controller that changes irrigation as the crop development stage and environmental conditions. Li et al. (2022) demonstrated that running AI inference at the Raspberry Pi level with NB-IoT can lessen delays and enhance ease of use a very important aspect for deployments with limited connectivity.

2.4 Technology Adoption

Per Zitan and Khalid (2021), Moroccan farmers' decision to use IoT irrigation is conditioned by their perception of the technology's usefulness, easiness of use, impact of social influence, and the degree to which they are aware of the environmental benefits. The main message is that while the technical aspect of development is important, it is not the only factor. Design that is easy to

use and farmer education are necessary components of the process. This is very important for Nepal, where many smallholders are mostly older people and digitally illiterate (Upadhyay et al., 2015).

2.5 Research Gaps and Project Justification

There are several gaps in the literature. Most systems have only been tested in controlled conditions; multi-season field trials in the smallholder contexts of South Asia are still rare. Further, the human elements of adoption, except for Zitan and Khalid (2021), remain largely untapped. The interoperability of different sensor platforms and protocols (LoRa, NB-IoT, ZigBee) is still an issue to be studied. Cybersecurity and data privacy issues in the agricultural IoT have hardly gotten any attention, even though AI is playing a bigger role in decision-making. This research intends to fill these gaps by developing a low-cost, open-standard system for Nepalese small farmers. The system is based on vendor-agnostic MQTT, JSON, and REST APIs features for interoperability. The integration of ESP32, LSTM inference, and solar power makes such a system feasible for the actual field situation in Nepal.

3 Project Aims, Objectives, and Scope

3.1 Research Questions

1. How far can an ESP32-based system be used for real-time measurement of soil and environmental parameters to aid irrigation decisions on Nepalese farms?
2. What is the level of accuracy of a machine learning model for predicting crop water requirements from sensor data, and what is the impact of this on water consumption compared to conventional practice?
3. How much can automated irrigation contribute to water use efficiency without sacrificing the reliability of supply under different weather conditions?

3.2 Project Aim

The goal is to design, build, and test a smart irrigation prototype by using/ IoT sensors and machine learning techniques/ to keep track of environmental conditions, predict water needs of crops, and minimize water wastage for sustainable small-scale farming in Nepal.

3.3 Project Objectives

Objective 1: The initial step would be to install a low power ESP32 microcontroller based IoT system which collects up-to-date data on soil moisture content, temperature, and humidity from a Nepalese-farm.

Objective 2: Using deep learning, build an LSTM model that can successfully forecast crop irrigation needs and the target model will be at least a certain threshold of Root

Mean Squared Error (RMSE) on a separate test set.

Objective 3: Cut down water usage by 30% or more via model-driven automatic irrigation control versus the regular methods based on the schedule of watering only.

Objective 4: Ensure the correct delivery of water on both timing and spatial distribution throughout the monitored ground area after irrigation.

Objective 5: Perform monitoring of the system reliability for the wet and dry season phases through different metrics like uptime, sensor accessibility, prediction precision, and so on.

3.4 Project Scope

The project involves hardware procurement and assembling (ESP32, soil moisture AHT10 BMP280 sensors), training data curation, LSTM development and validation, cloud dashboard implementation, and performance testing over at least two weather seasons.

It does not cover multi-farm deployment, satellite integration, cybersecurity hardening, and socio-economic adoption analysis — all considered as future work.

4 Project Design

4.1 System Architecture

The system follows a hardware–software co-design approach organised into five layers — sensing, edge processing, AI inference, communication, and actuation — illustrated in Figure 1.

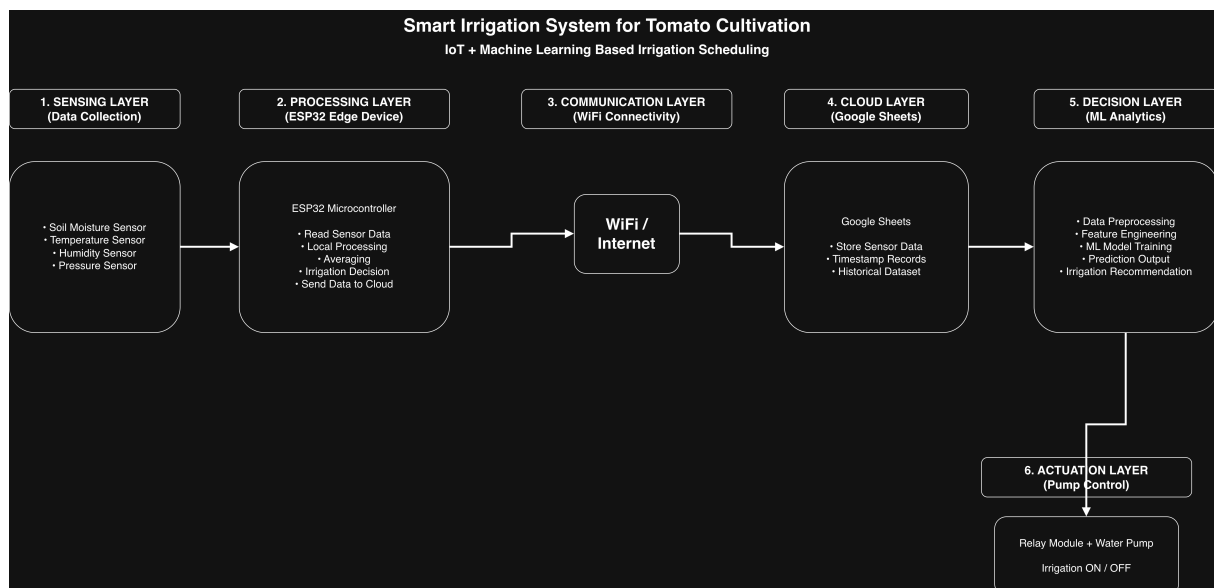


Figure 1: System architecture of the proposed smart irrigation system.

Sensing Layer. Sensors for soil moisture temperature humidity and light intensity are interfaced with an ESP32 microcontroller that was chosen for its dual cores, integrated Wi-Fi and Bluetooth as well as Really it has been widely used in agricultural IoT (Singha et al., 2025). Samples are taken every five minutes to maintain the right tradeoff between data quality and power consumption. Solar energy harvesting at 30 W/m² can provide around 12 hours of independent operation time after the main battery has been drained.

Edge Processing Layer. The ESP32 performs noise filtering, anomaly detection, and local computation of derived metrics like vapor pressure deficit (VPD) which reduces the amount of data sent to the cloud and allows local decision-making in case of no connectivity.

AI / Decision Layer. An LSTM network predicts crop water needs for the next 24 hours based on pre-processed sensor series and historical weather data as described by Meng et al. (2023). The final model is converted to TensorFlow Lite so that it can be run on the ESP32 locally without the internet.

Communication Layer. Data is sent through MQTT over Wi-Fi to a cloud broker (Mosquitto running on AWS IoT Core). In case of weak-signal sites, LoRa can be used as a long-range fallback. A web dashboard built in React. js/Node. js is used for remote monitoring and manual control.

Actuation Layer. Relay modules switch solenoid valves on water supply lines following model outputs. The whole field unit is powered off-grid by a solar panel with MPPT controller and lithium battery backup Sohan and Nahar (2022).

4.2 Training Data Strategy

Model training proceeds in two phases.

Phase 1 (pre-deployment) starts the process of model training and identifies the model architecture based on the publicly available agricultural datasets from UCI and Kaggle that are similar in climate to Nepal's Terai and hill regions (Meng et al., 2023).

Phase 2 (in-field refinement) After the sensor network has been installed and is functioning, adjusts the model using the data collected locally, resulting in better calibration of the model to the specific characteristics of the soil, crops, and microclimate of the site. The performance of Phase 1 and Phase 2 models will be evaluated against each other to measure how much local adaptation contributes to the model.

4.3 Evaluation Framework

Five metrics will be used to assess whether objectives are met:

M1: Prediction accuracy: RMSE and MAE on a held-out 20% test split (Objective 2).

M2: Water saving: Volume applied by the automated system versus the conventional site baseline; target 30% reduction (Objective 3) (Singha et al., 2025).

M3: System uptime: Percentage of scheduled readings that were done successfully; target 95% per deployment phase; failures will be analysed by type.

M4: Response latency: Time in seconds from model command to opening of valve; target is under 10 seconds.

M5: Cross-season generalisation: All metrics will be reported separately for the wet and dry seasons (Objective 5).

In case it is appropriate, results will be analysed through descriptive statistics and paired t-tests on daily water usage, and qualitative observations from pilot-site farmers will also be used as supplementary resources.

4.4 Justification of Methods

The ESP32 was chosen as the best fit when compared to Arduino Uno (which has very limited memory for TFLite) and Raspberry Pi (which consumes more power than what a solar panel can provide). It features a 240 MHz dual-core CPU, built-in wireless connectivity, and very low idle current.

Besides, MQTT was opted over HTTP REST because of its lightweight publishsubscribe pattern, and smaller packet overhead that are more reliable, even on the intermittent bandwidth typical of rural Nepal.

4.5 Project Timeline

The project will be carried out in 30 weeks through 6 phases. Some main milestones are: a working prototype of the hardware ready (week 10); interim paper handed in (week 14); system integration done (week 20); final submission (week 30). Figure 2 depicts the entire timeline. Figure 2 shows the full schedule.

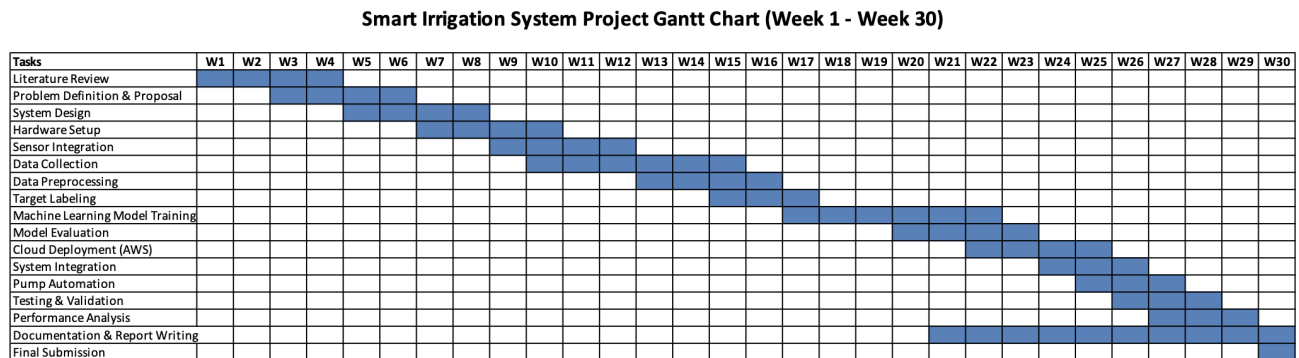


Figure 2: Project timeline (Gantt chart).

5 Feasibility, Risks, and Ethical Aspects

5.1 Feasibility

The project can be realized both technically and financially within the given time and budget constraints. ESP32 and the selected sensors (AHT10 BMP280 soil moisture) are not only affordable but also have extensive documentation. TensorFlow Lite and MQTT don't cost a thing and are open for everyone. Using the cloud is still within the limits of the AWS IoT Core free tier. The solar setup repeats a proven design (Sohan and Nahar, 2022), and a background in embedded systems and Python-based machine learning lessens the risk of needing to learn new skills.

5.2 Risk Analysis and Mitigation

Table 1 presents the principal risks and mitigation strategies.

5.3 Ethical Considerations

Despite never collecting clinical and sensitive personal data, the obligations below are recognised:

- **Farmer involvement:** Informed consent is required in writing from all landowner. Setting up field tests. Participants' data will be kept anonymous and used only for research.
- **Data protection:** Sensor data will be stored on encrypted cloud servers. Private data will not be stored nor available to other users. These data will be gathered.

Risk	Likelihood (L/M/H)	Impact (L/M/H)	Mitigation
Hardware procurement delay	<i>Medium</i>	<i>High</i>	Order early; identify local Nepalese suppliers as backup; start development with simulated data.
Poor Wi-Fi at field site	<i>High</i>	<i>Medium</i>	Deploy LoRa as fallback; confirm that edge-only LSTM inference can work without cloud connectivity.
Insufficient training data	<i>Medium</i>	<i>High</i>	Use UCI/Kaggle datasets for Phase 1 (Section 4.2); And, local data will be used to do refinements in Phase 2.
TFLite/ESP32 integration issues	<i>Medium</i>	<i>Medium</i>	Allocate dedicated integration time (Weeks 12–14); engage supervisor and community resources.

Table 1: Risk assessment and mitigation strategies.

- **Environmental responsibility:** The system minimizes water consumptions and causes none to the environment. Chemicals or toxic substances; solar power reduces the emission of carbon dioxide.
- **Socio-economic impact:** Water and yield gains derive directly from the sustainability of smallholders.

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